BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Characters and Images

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Convert hexadecimal A3 to eight-bit binary.

2. How many different codes can 7 bits represent?
   A) 7
   B) 64
   C) 128
   D) 256

3. "Metadata is another name for the colour of every pixel." Correct this. Explain why.

4. (a) A bitmap uses 5 bits per pixel. How many colours can it show? What if you add one bit? (b) A 50 x 20 pixel image doubles in width and height. Colour depth stays the same. How do pixel count and raw size change? Explain.

Today's objectives

Recall the essential ideas of characters and images. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Character set
Mapping between characters and numerical codes.

ASCII
A character encoding standard with a relatively limited basic character range.

Unicode
A universal standard covering many characters and writing systems.

Resolution
Width x height measured in pixels.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Characters need codes
Character sets map symbols to numerical codes stored in binary; ASCII and Unicode are examples.

Pixels build bitmaps
A bitmap stores the colour of each pixel in a grid. Resolution is width in pixels x height in pixels.

Colour depth
Colour depth is the number of bits per pixel. More bits allow more possible colours.

Image file size
Uncompressed size (bits) = width x height x colour depth; divide by eight for bytes (ignoring metadata).

I Do - Worked model

Calculate the uncompressed size of a 200 x 100 pixel bitmap with 8-bit colour depth. [4 marks]

- Identify width and height
- 200 x 100 = 20,000 pixels.
- Multiply by colour depth: 20,000 x 8 = 160,000 bits.
- Divide by eight: 20,000 bytes. [4]

We Do - Guided practice

1. A 100 x 80 image uses 16-bit colour. Calculate its uncompressed size in bytes. [4]

2. Explain the impact of increasing colour depth from 8 to 16 bits. [4]

3. Explain one difference between ASCII and Unicode. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Calculate the raw image data in bytes.

```
width = 10
height = 8
bitsPerPixel = 4
bytes = ...
print(bytes)
```

You Do - Independent exam questions

Q2. An image is 50 pixels wide and 40 pixels high, using 8 bits per pixel. Calculate the uncompressed image size in bytes. [4]

Q3. Explain why increasing an image's resolution increases file size when colour depth stays constant. [3]

Q4. State one advantage of Unicode compared with basic ASCII. [2]

Exit Ticket - True or False?

A trace table can help check how values change during a program.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 10100011: A is 1010 and 3 is 0011.

2. C - 128

3. Extra image details, such as width, height or colour depth. These are separate from pixel data.

4. (a) 32 colours; one extra bit doubles this to 64. (b) 1,000 pixels become 4,000. Raw size is four times larger: both dimensions double.

I Do - Explain the model

Identify width and height

200 x 100 = 20,000 pixels.

Multiply by colour depth: 20,000 x 8 = 160,000 bits.

Divide by eight: 20,000 bytes. [4]

We Do - Model answers

1. 100 x 80=8000 pixels [1]; x 16=128,000 bits [1]; / 8 [1] = 16,000 bytes [1].

2. More possible colours per pixel [1] allows richer colour representation [1], but doubles uncompressed bits per pixel [1] and increases file size at equal resolution [1].

3. ASCII is a smaller character set [1]; Unicode supports a much wider range of characters and writing systems [1], improving international text support [1].

Code First - Answer

```
width = 10
height = 8
bitsPerPixel = 4
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

Expected output: 40

You Do - Self-mark

Q2: 50 x 40=2000 pixels [1]; x 8=16,000 bits [1]; / 8 [1] =2000 bytes [1].

Q3: More pixels are stored [1]; each pixel still requires the same number of bits [1]; total uncompressed bits increase [1].

Q4: Unicode represents a wider range of characters/scripts [1], supporting multilingual text [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.